

IYAD KHATTAB

iyadahmedkhattab@gmail.com | 201005040210 | New Cairo, Egypt | github.com/iyadkhattab | linkedin.com/in/iyad-khattab

Computer Engineering student at the German University in Cairo (GUC) with a 1.38 GPA (A-), hands-on experience in software development, machine learning, data analysis, and AI systems. Proficient in Java, Python, C, JavaScript, SQL, and Flutter, with experience building full-stack applications, ML pipelines, and AI-powered automation systems. Interested in Software Engineering, Data Science, Machine Learning, and Agentic AI.

EDUCATION

B.Sc. in Computer Engineering (MET)

German University in Cairo (GUC)

Sep 2023 – Present

- GPA: 1.38(A-) | **Relevant Coursework:** Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Software Engineering, Operating Systems (Processes, Threads, System Calls, Memory Management, Scheduling), Computer Networks, Computer Architecture, Probability & Statistics, Digital System Design, Digital Logic Design.

EXPERIENCE

Junior Teaching Assistant — CS1 (Python) & CS2 (Java)

GUC

Feb 2025 – Dec 2025

- Conducted programming lab sessions and supported students in **Python and Java**, covering object-oriented programming, algorithms, debugging, and problem solving
- Guided students through programming assignments and technical challenges, strengthening their understanding of **data structures, algorithms, and software development principles**.
- Facilitated technical discussions and provided hands-on assistance with debugging and implementation

CIB Future Vest Internship — "The Green Leap"

Commercial International Bank

Jul – Aug 2025

- Selected competitively for CIB's flagship sustainable finance program focused on ESG principles.
- Received SAS training on advanced data analytics, AI tools, and agentic AI systems for ESG reporting.
- Earned completion certificate after online exam and daily assignments.

IEEE GUC — Marketing Committee Member

Oct 2024 – Present

- Supported digital marketing campaigns and promotional activities for **technical workshops, student events, and engineering initiatives**.

Canon Medical Equipment Center — Intern

New Cairo

Jul 2024

- Gained hands-on experience with X-ray and ultrasound equipment.
- Supported technical staff in identifying equipment issues and understanding system operation.

PROJECTS & CERTIFICATES

AI Email Assistant — Agentic AI & Full-Stack Application

Python, FastAPI, Streamlit, GmailAPI

- Built an AI-powered email assistant that **retrieves, classifies, summarizes, and assists with responses to Gmail messages**.
- Developed a **FastAPI backend** with modular services for authentication, Gmail integration, email processing, AI classification, and response generation.
- Built a **Streamlit frontend** for email triage, summaries, and AI-generated reply drafts.
- Integrated **Gmail OAuth 2.0** authentication and an AI provider architecture supporting configurable language models.
- Designed the system as an AI-driven workflow for **email prioritization, summarization, decision support, and automated response generation**.

Supermarket Shopping App

Flutter/Dart — Microsoft × Sprints 2025

- Built a fully functional cross-platform mobile app (Android & iOS) with product listing, cart, search, and filter.
- Worked with Firebase, Git, Android Studio; applied UX/UI and clean code principles.

Heart Disease Prediction Project

Python, Pandas, Numpy, Scikit-learn, Matplotlib, PyMC — Microsoft × Sprints 2025

- Developed an end-to-end **machine learning pipeline** for heart disease prediction, including data preprocessing, feature selection, model training, and evaluation.
- Implemented and compared **Logistic Regression, Random Forest, Support Vector Machine (SVM), and K-Means** models.
- Developed probabilistic risk prediction using **PyMC and NUTS sampling**.
- Used **Pandas, NumPy, Matplotlib, and Scikit-learn** for data analysis, visualization, and model development.

Operating System Scheduler Simulator

Java — Operating System

- Built a simulated operating system implementing **process management, system calls, memory management, PCBs, mutexes, process synchronization, and disk swapping**.
- Implemented **HRRN, Round Robin, and Multi-Level Feedback Queue scheduling**, with Ready/Blocked queues and process state management.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, Dart, VHDL

Testing & QA: Unit Testing, JUnit, Test Automation, Debugging

Frameworks: Flutter, FastAPI, Node.js, Streamlit, Firebase

Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib, PyMC, Machine Learning

Databases: SQL, MongoDB, SQLAlchemy

Tools: Git, GitHub, SAS Viya, Android Studio, VS Code, ModelSim

CS: OOP, Data Structures & Algorithms, Software Engineering, Operating Systems, Computer Networks

Languages Spoken: Arabic (Native), English (Fluent), German (A2)